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Hardek

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(54) **DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS**

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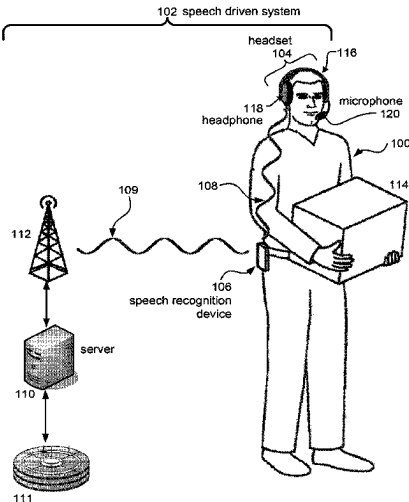
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(57) **ABSTRACT**

A device, system, and method whereby a speech-driven system can distinguish speech obtained from users of the system from other speech spoken by background persons, as well as from background speech from public address systems. In one aspect, the present system and method prepares, in advance of field-use, a voice-data file which is created in a training environment. The training environment exhibits both desired user speech and unwanted background speech, including unwanted speech from persons other than a user and also speech from a PA system. The speech recognition system is trained or otherwise programmed to identify wanted user speech which may be spoken concurrently with the background sounds. In an embodiment, during the pre-field-use phase the training or programming may be accomplished by having persons who are training listeners audit the pre-recorded sounds to identify the desired user speech. A processor-based learning system is trained to duplicate the assessments made by the human listeners.

18 Claims, 6 Drawing Sheets



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continuation of application No. 17/449,213, filed on Sep. 28, 2021, now Pat. No. 11,837,253, which is a continuation of application No. 16/695,555, filed on Nov. 26, 2019, now Pat. No. 11,158,336, which is a continuation of application No. 15/220,584, filed on Jul. 27, 2016, now Pat. No. 10,714,121.

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FIG. 1

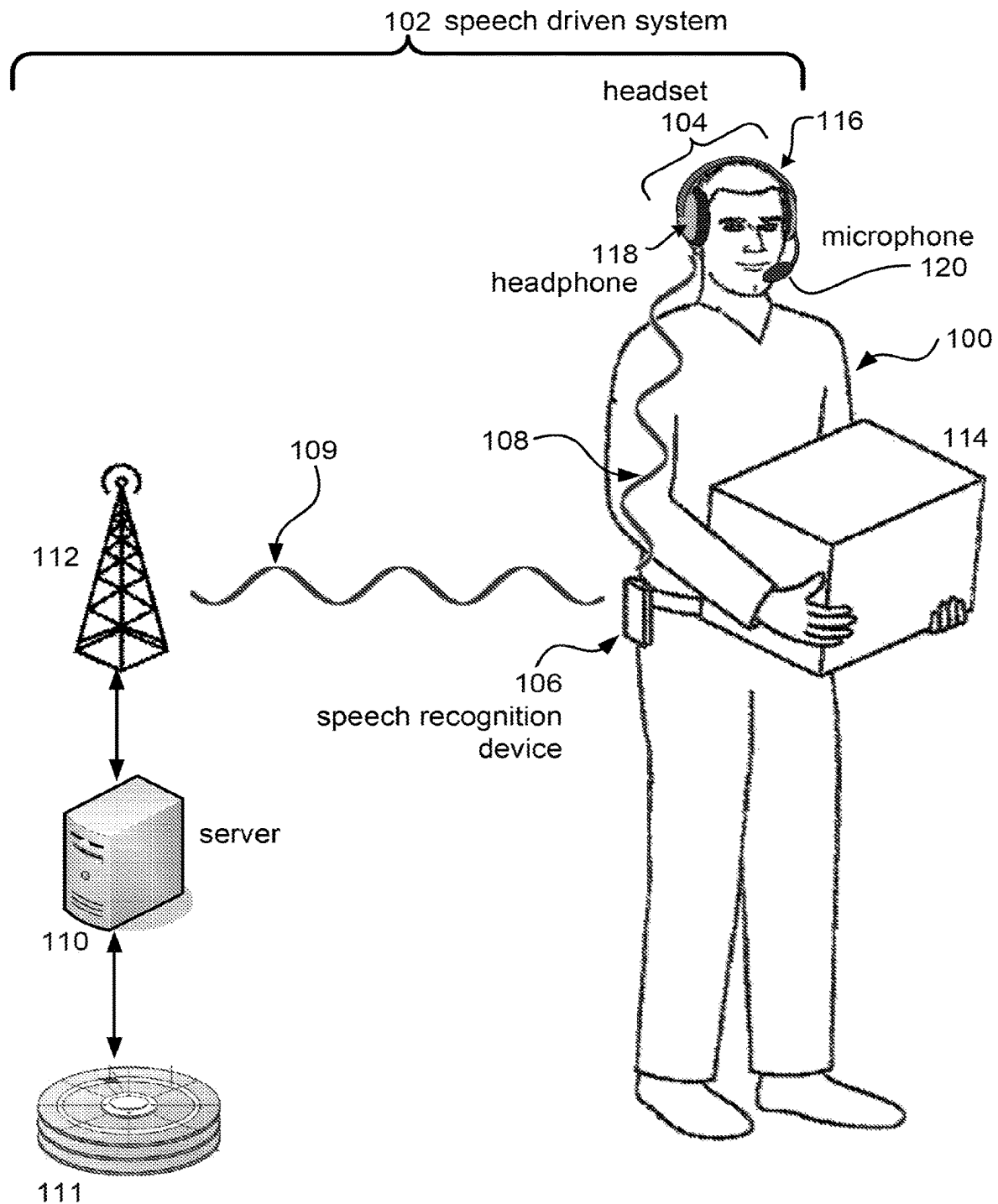


FIG. 2

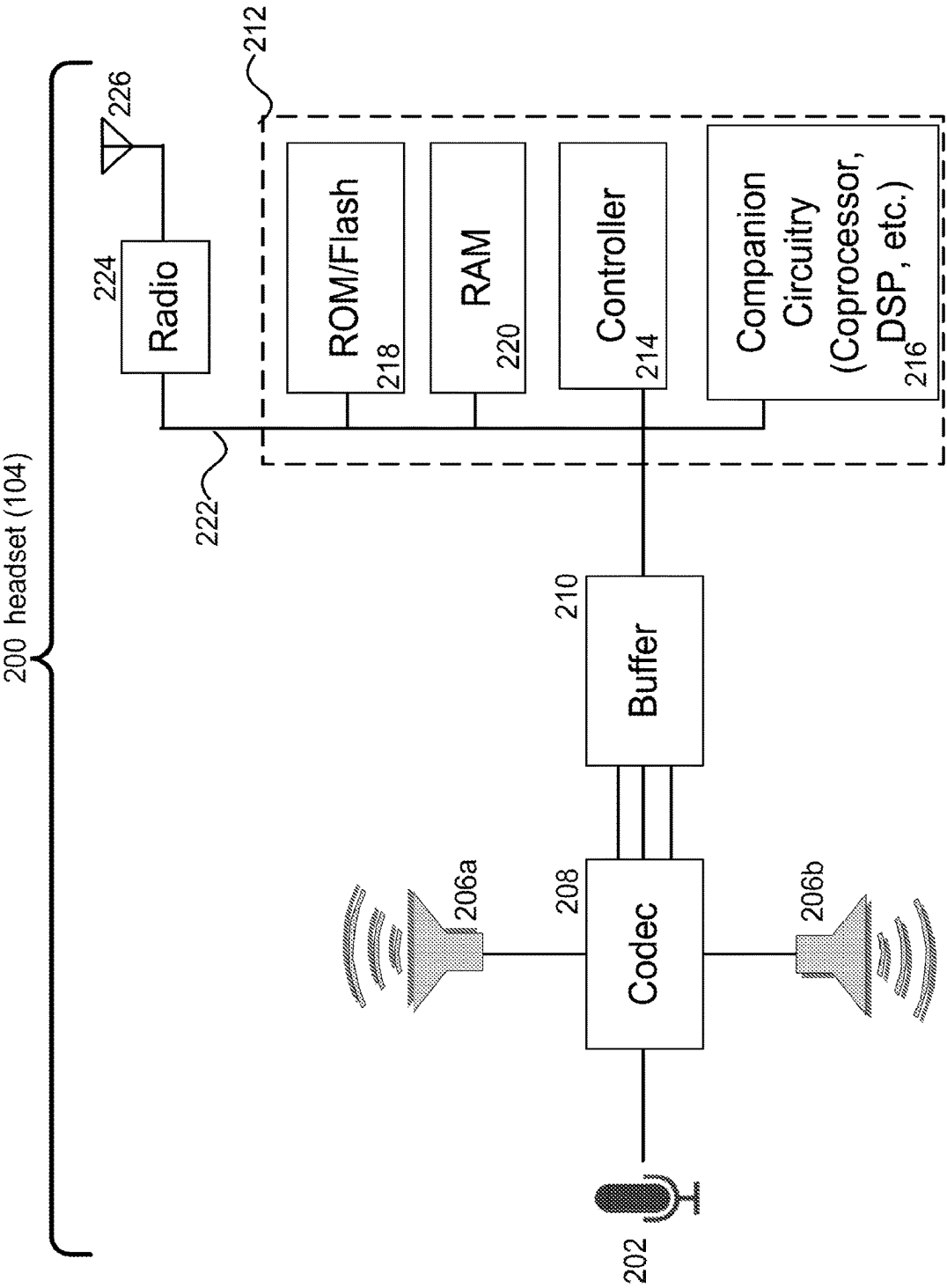


FIG. 3

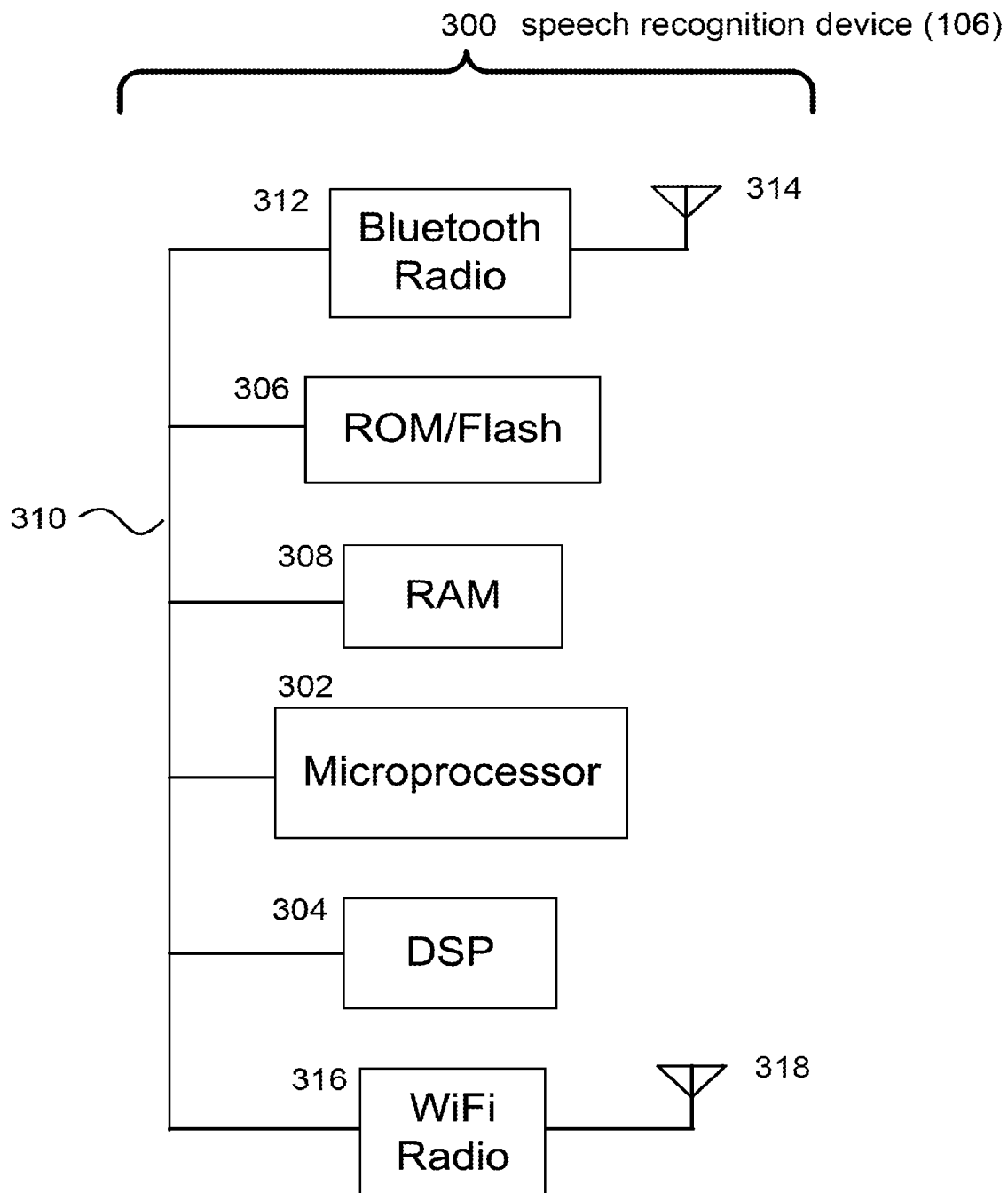


FIG. 4

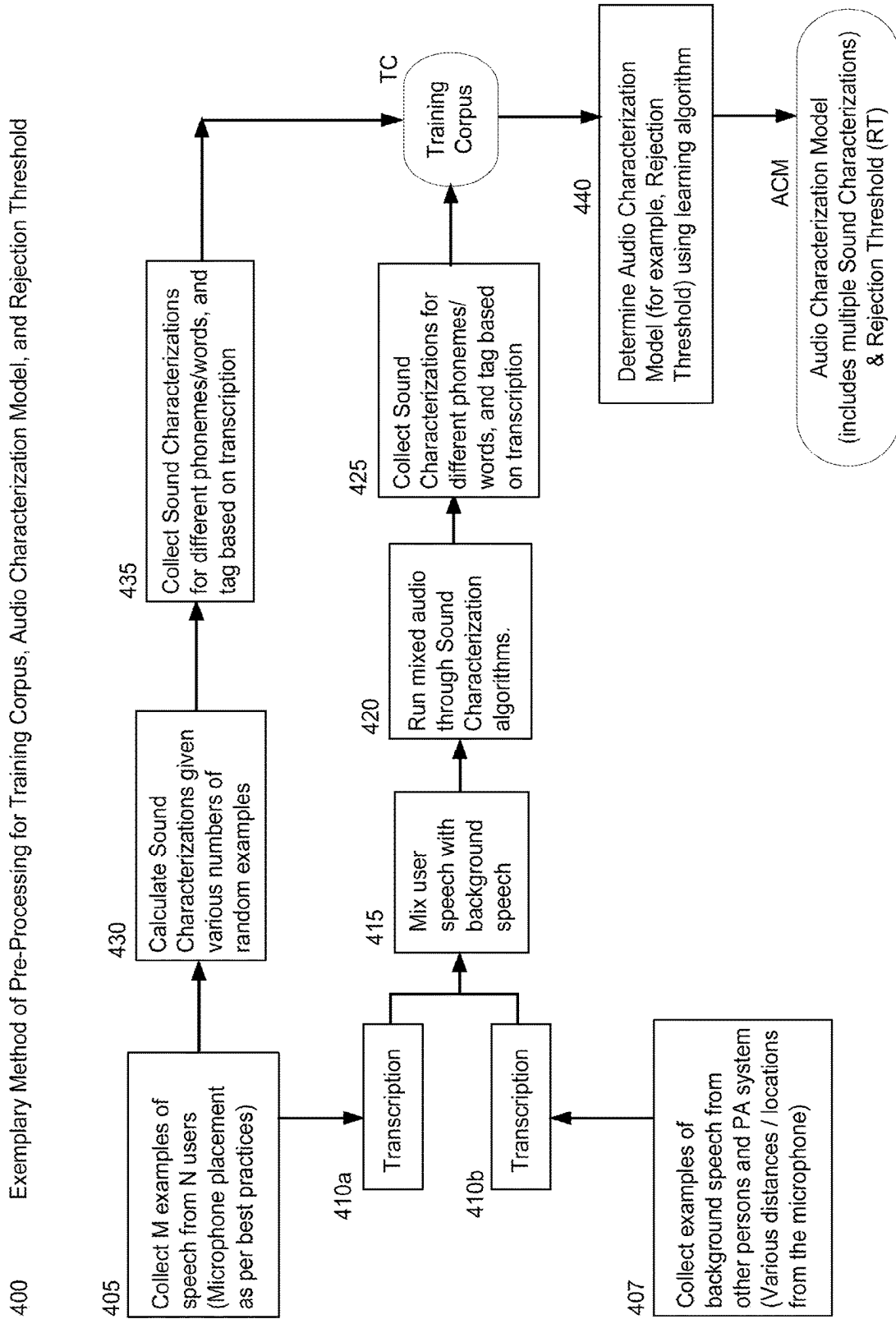
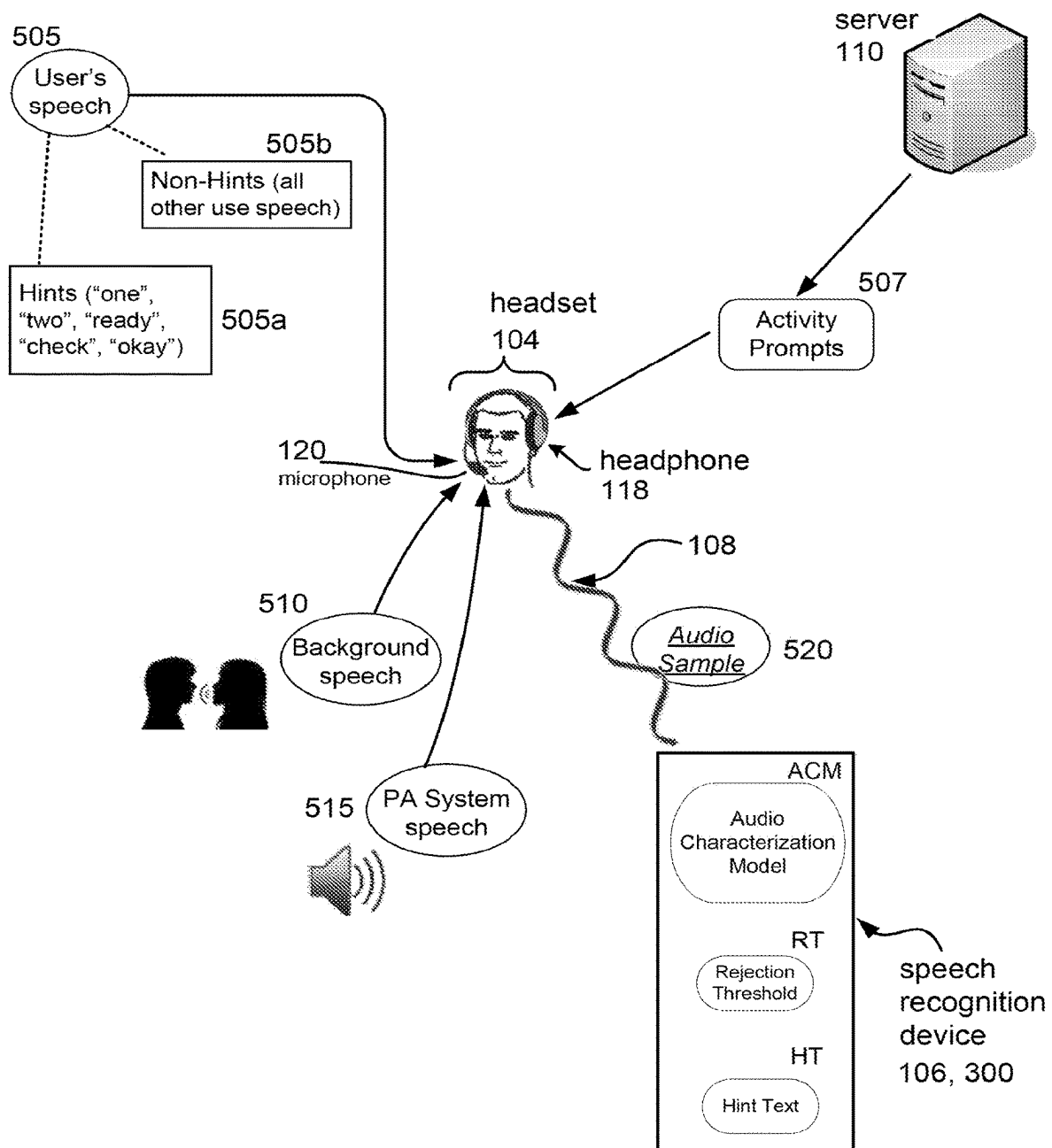


FIG. 5

500 Field-Use Audio Environment



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DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of and claims priority to U.S. application Ser. No. 18/328,034, filed Jun. 2, 2023, titled “DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS,” which is a continuation of and claims priority to U.S. application Ser. No. 17/449,213, filed Sep. 28, 2021, titled “DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS,” which is a continuation of and claims priority to U.S. application Ser. No. 16/695,555, filed Nov. 26, 2019, titled “DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS,” (now U.S. Pat. No. 11,158,336), which is a continuation of and claims priority to U.S. application Ser. No. 15/220,584, filed Jul. 27, 2016, titled “DISTINGUISHING USER SPEECH FROM BACKGROUND SPEECH IN SPEECH-DENSE ENVIRONMENTS,” (now U.S. Pat. No. 10,714,121), the contents of which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

The present invention relates to a method and apparatus for recognition of human speech, and more particularly, to a method and apparatus to distinguish user speech which is the desired focus of machine-interpretation from extraneous background speech.

BACKGROUND

In modern production environments, it is increasingly desirable for human operators to be able to record data and to control electronic devices in a “hands-free” mode, typically via speech control. This typically entails the use of portable electronic voice-processing devices which can detect human speech, interpret the speech, and process the speech to recognize words, to record data, and/or to control nearby electronic systems.

Voice-driven systems typically include at least one microphone and at least one processor-based device (e.g., computer system) which is operated in response to human voice or spoken input, for instance spoken commands and/or spoken information.

There are numerous applications in which voice-driven systems may be employed. For instance, there are many applications where it is advantageous for a user to have their hands free to perform tasks other than operating a keyboard, keypad, mouse, trackball or other user input device. An example of one such application is a warehouse, where a user may need to handle items such as boxes while concurrently interacting with a processor-based device. Another example application is a courier or delivery person, who may be handling parcels or driving a vehicle while concurrently interacting with a processor-based device. An example of a further such application is a medical care provider, who may be using their hands during the performance of therapeutic or diagnostic medical services, while concurrently interacting with a processor-based device. There are of course numerous other examples of applications.

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In many of these exemplary applications it is also advantageous or even necessary for the user to be mobile. For applications in which mobility is desirable, the user may wear a headset and a portable processor-based device (referred to below in this document at the speech recognition device **106**, **300**, or SRD). The headset typically includes at least one loud-speaker and/or microphone. The portable processor-based device typically takes the form of a wearable computer system. The headset is communicatively coupled to the portable processor-based device, for instance via a coiled wire or a wireless connection, for example, a Bluetooth connection. In some embodiments, the portable processor-based device may be incorporated directly into the headset.

In some applications, the portable processor-based device may in turn be communicatively coupled to a host or backend computer system (e.g., server computer). In many applications, two or more portable processor-based devices (clients) may be communicatively coupled to the host or backend computer system/server.

The server may function as a centralized computer system providing computing and data-processing functions to various users via respective portable processor-based devices and headsets. Such may, for example, be advantageously employed in an inventory management system in which a central or server computer system performs tracking and management; a plurality of users each wearing respective portable computer systems and headsets interface with the central or server computer system.

This client (headset)/server approach allows the user(s) to receive audible instructions and/or information from the server of the voice driven system. For instance, the user may: receive from the server voice instructions; may ask questions of the server; may provide to the server reports on progress of their assigned tasks; and may also report working conditions, such as inventory shortages, damaged goods or parcels; and/or the user may receive directions such as location information specifying locations for picking up or delivering goods.

Background Sounds

Voice driven systems are often utilized in noisy environments where various extraneous sounds interfere with voice or spoken input. For example, in a warehouse or logistics center environment, extraneous sounds are often prevalent, including for instance: public address announcements; conversations from persons which are not intended as input (that is, persons other than the user of the voice driven system); and/or the movement of boxes or pallets; noise from the operation of lift vehicles (e.g., forklifts), motors, compressors, and other nearby machinery. To be effective, voice driven systems need to distinguish between voice or speech as intended input and extraneous background sounds, including unwanted voices, which may otherwise be erroneously interpreted as desired speech from a headset-wearing user.

Sounds or noise associated with public address (PA) systems are particularly difficult to address. Public address systems are intentionally loud, so that announcements can be heard above other extraneous noise in the ambient environment. Therefore, it is very likely that a headset microphone will pick up such sounds. Additionally, public address system announcements are not unintelligible noise, but rather are typically human voice or spoken, thereby having many of the same aural qualities as voice or spoken input.

Therefore, there exists a need for a system and method for addressing extraneous sounds including background speech

and PA system speech, in order to prevent those extraneous sounds from interfering with the desired operation of the voice driven systems.

SUMMARY

Accordingly, in one aspect, the present system and method solves the problem by preparing, in advance of field-use, a voice-data model which is created in a training environment, where the training environment exhibits both desired user speech and unwanted background sounds, including unwanted speech from persons other than a user, and also unwanted speech from a PA system.

The speech recognition system is trained or otherwise programmed to identify wanted user speech which may be spoken concurrently with the background sounds. In an embodiment, during the pre-field-use phase the training or programming is accomplished in part by having persons who are training listeners audit the pre-recorded sounds, and having the training listeners identify the desired user speech, a process referred to as “tagging”. Tagging may also entail have the training listeners identify background speech from persons other than the user, background speech from PA system sounds, and other environmental noises. In an embodiment, during the pre-field-use phase the training or programming is further accomplished in part by training a processor-based learning system to duplicate the assessments made by the human listeners.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a view of an exemplary speech-driven system according to one exemplary embodiment of the present system and method.

FIG. 2 is a system diagram of a headset identical or similar to that of FIG. 1, according to one exemplary embodiment of the present system and method.

FIG. 3 is a system view of a speech recognition device identical or similar to that of FIG. 1, according to one exemplary embodiment of the present system and method.

FIG. 4 is a flow-chart of an exemplary method for pre-field-use creation of a training corpus, audio characterization model, and rejection threshold for an exemplary speech-driven system, according to one embodiment of the present system and method.

FIG. 5 illustrates an exemplary audio-environment for a deployment and field-use of an exemplary headset and exemplary speech recognition device according to an embodiment of the present system and method.

FIG. 6 is a flow-chart of an exemplary method of field-use of an exemplary speech recognition device for distinguishing user speech from background speech.

DETAILED DESCRIPTION

In the following description, certain specific details are set forth in order to provide a thorough understanding of various embodiments. However, one skilled in the art will understand that the invention may be practiced without these details. In other instances, well-known structures associated with voice recognition systems and speech recognition devices have not been shown or described in detail to avoid unnecessarily obscuring descriptions of the embodiments.

Unless the context requires otherwise, throughout the specification and claims which follow, the word “comprise” and variations thereof, such as, “comprises” and “compris-

ing” are to be construed in an open sense, that is as “including, but not limited to.”

Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

The headings provided herein are for convenience only and do not interpret the scope or meaning of the claimed invention.

Electronic System for Voice Processing

The present system and method embraces electronic devices designed to interpret human speech and language, and to operate in response to human speech, also known as voice-driven systems, speech-driven systems, or spoken-language recognition systems.

FIG. 1 shows a user 100 interacting with an exemplary speech driven system 102, according to one embodiment of the present system and method.

In particular, the speech driven system 102 includes a headset 104 and a processor-based speech recognition device 106. In use, the user typically wears the headset 104, and optionally wears the processor-based speech recognition device 106. The processor-based speech recognition device 106 is communicatively coupled, either directly or indirectly (that is, via either wired or wireless coupling), with the headset 104. For example, the processor-based speech recognition device 106 and headset 104 may be wirelessly communicatively coupled via one or more radios (e.g., transmitters, receivers, transceivers) as indicated by radio frequency signal 108. Alternatively, the processor-based speech recognition device 106 and headset 104 may be communicatively coupled via one or more cables, for instance one or more wire or optical cables (not shown).

Optionally, the speech driven system 102 may also include one or more backend computer systems 110 (only one shown), which may include or be communicatively coupled to one or more data stores stored on one or more non-transitory computer- or processor-readable media 111. The backend computer system(s) 110 is or are communicatively coupled to one or more processor-based speech recognition devices 106. For example, a wireless networking system may include one or more antennas 112 (only one shown) positioned about a work environment. Antenna 112 can provide wireless communications (for example, by radio frequency signal 109) between the one or more processor-based speech recognition devices 106 and the one or more backend computer system(s) 110.

The user 100 may engage in various activities which may require the use of the user’s hands, for instance to handle goods or packages 114. Alternatively, the activities may not require use of the user’s hands; however hand-free operation may be more comfortable or otherwise advantageous to the user 100.

The headset 104 may include a headband 116, one or more loud-speakers or headphones 118 (only one visible in FIG. 1), one or more microphones 120 (one visible in FIG. 1), and internal circuitry (not illustrated). The headband 116 allows the headset 104 to be securely worn by the user 100, and positions the loud-speakers 118 at least proximate one ear or next to each ear of the user 100. The microphone 120

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may be positioned proximate and oriented toward a mouth of the user **100** when the headset **104** is worn.

The circuitry (not shown in FIG. **1**) of the headset **104** may incorporate audio processing circuits such as audio filters and correlation circuitry associated with speech detection and/or speech recognition.

The processor-based speech recognition device **106** may be portable or stationary. For example, the processor-based speech recognition device **106** may be worn by the user **100**, for instance on a belt as illustrated in FIG. **1**. This allows the headset **104** to use relatively short range wireless communications devices, for instance Bluetooth radios, while ensuring that communications between the headset **104** and the processor-based speech recognition devices **106** is maintained during normal use.

Alternatively, the processor-based speech recognition device **106** may be manually carried or otherwise transported, for instance on a vehicle (e.g., fork lift, tug). Alternatively or additionally, the processor-based speech recognition device **106** may be stationary. Such implementations may employ a plurality of antennas positioned throughout a work environment and/or sufficiently more powerful communications devices, for instance WiFi radios.

The circuitry (not shown in FIG. **1**) of the processor-based speech recognition device **106** may incorporate audio processing circuits for tasks such noise suppression and modeling, features vector generation, decoding, and other circuitry associated with speech detection and/or speech recognition.

The headset **104** and processor-based speech recognition device **106** permit various users **100** to communicate with one or more backend computer systems **110** (e.g., server computer systems). In use, the processor-based speech recognition device **106** receives digital instructions from the backend computer system **110** and converts those instructions to audio, which is provided to the user **100** via loud-speakers **118** of the headset **104**. The user **100** provides spoken input via the microphone **120** of the headset, which the processor-based speech recognition device **106** may convert to a digital format (e.g., words, text, or encoding symbolic of words and text) to be transferred to the backend computer system **110**.

The backend computer system(s) **110** may be part of a larger system for sending and receiving information regarding the activities and tasks to be performed by the user(s) **100**. The backend computer system(s) **110** may execute one or more system software routines, programs or packages for handling particular tasks. Tasks may, for example, include tasks related to inventory and warehouse management.

In an alternative embodiment of the present system and method, the backend computer system(s) **110** may implement some, or all, of the functionality otherwise described herein as being associated with the processor-based speech recognition device **106**.

The backend computer system/server **110** may be any targeted computer or automated device, and may be located anywhere with respect to the user and the various components. For instance, the backend computer system **110** will typically be located remotely from the user, such as in another room or facility.

However, the background computer system **110** may be located locally with the user, for instance carried or worn by the user or carried by a vehicle operated by the user. In some implementations, that backend computer system **110** may be combined with the processor-based speech recognition device **106**.

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In an alternative embodiment, the headset **104** and the speech recognition device (SRD) **106** may be connected and may communicate via a wired connection, such as a coiled cable.

5 Headset

FIG. **2** shows some of the components of an exemplary headset **200**, according to one exemplary embodiment of the present system and method. The headset **200** may be similar or even identical to the exemplary headset **104** of FIG. **1**.

10 The headset **200** includes a microphone **202**, and may include one or more secondary microphones (not shown). The microphone **202** is operable as a transducer to convert acoustic energy (e.g., sounds, such as voice or other sounds) to analog signals (e.g., voltages, currents) that have respective signal levels. The headset **200** preferably includes one or more loudspeakers **206a**, **206b** (two shown, collectively **206**). Each of the loud-speakers **206** is operable as a transducer to convert analog signals (e.g., voltages, currents) that have respective signal levels into acoustic energy (e.g., sounds, such as recorded or artificially generated spoken syllables, words or phrases or utterances).

15 The microphone(s) **202** is (are) positioned or configured (e.g., directional and oriented) to primarily capture speech or utterances by the user **100**. However, the microphone **202** may also capture background speech from other users in the work environment, as well as background speech from PA systems. In this document, background speech will be understood to include both speech from persons other than the user **100** and Public Address (PA) system speech.

20 The microphone **202** may be positioned such that when the headset **104** (FIG. **1**) is worn by a user **100**, the microphone **202** is positioned close to the mouth of the user **100**. For example, the microphone **202** may be carried at an end of an arm/boom of the headset **104** (FIG. **1**), positioning the primary microphone **202** proximate to the mouth of the user **100**. Consequently, the speech sounds or utterances by the user **100** are typically louder, as recorded at the microphone **202**, than background speech sounds from other persons who are some distance from the microphone **202**.

25 With respect to PA systems, background speech from a PA system may be amplified, and so may be picked up by the microphone **202** as being approximately as loud as the user speech. However, due to various factors—emanating from a remote loud-speaker, frequency band limitations of the PA system, and due to echoes and other factors—remote speech from a PA system may have different acoustic qualities at the microphone **202**, as compared to the acoustic qualities of user speech.

30 In other words, user speech or other utterances by the user **100** are likely to have different acoustic signatures than background speech from other persons at some distance from the user **100**, or and also different acoustic signatures from sounds from a PA system. In one embodiment, the present system and method may rely, in part or in whole, on signal processing techniques, as applied to such acoustic differences, to distinguish user speech from background speech.

35 In an alternative embodiment, some implementations of the present system and method may employ additional secondary microphones (not shown), for example two or more secondary microphones, to help distinguish user speech from background speech.

The headset **200** may include one or more audio coder/decoders (CODECs). For example, the headset **200** may include an audio CODEC **208** coupled to the microphone(s) **202** to process analog signals from the microphone **202** and produce digital signals representative of the analog signals.

The CODEC **208** or another audio CODEC (not shown) may be coupled to the one or more loud-speakers **206** to produce analog drive signals from digital signals in order to drive the loudspeakers **206**. Suitable audio CODECs may for example include the audio CODEC commercially available from Philips under the identifier UDA **1341** and other similar audio CODECs.

The headset **200** may include one or more buffers **210**. The buffer(s) **210** may temporarily store or hold signals. The buffer **210** is illustrated as positioned relatively downstream of the CODEC **208** in a signal flow from the microphone **202**.

The headset **200** includes a control subsystem **212**. The control subsystem **212** may, for example include one or more controllers **214**, one or more sets of companion circuitry **216**, and one or more non-transitory computer- or processor-readable storage media such as a non-volatile memory **218** and volatile memory **220**.

The controller(s) **214** may take a variety of forms, for instance one or more microcontrollers, microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), programmable gate arrays (PGAs), graphical processing unit (CPUs) and/or programmable logic controllers (PLCs). The controller(s) **214** may, for example, take the form of a processor commercially available from CSR under the identifier BlueCore5 Multimedia. The BlueCore5 Multimedia does not require companion circuitry **216**. Alternatively, the controller(s) **214** may take the form of a processor commercially available from Intel under the identifier SA-1110. Optional companion circuitry **216** may take the form of one or more digital, or optionally analog, circuits, which may, or may not, be in the form of one or more integrated circuits. The companion circuitry **216** may, for example, take the form of a companion chip commercially available from Intel under the identifier SA-1111. The controller(s) **214** may function as a main processor, with the companion circuitry functioning as a co-processor to handle specific tasks. In some implementations, the companion circuitry **216** may take the form of one or more DSPs or GPUs.

Non-volatile memory **218** may take a variety of forms, for example one or more read only memories (ROMs), one or more writeable memories, for instance EEPROM and/or one or more FLASH memories. The volatile memory **220** may take a variety of forms, for example one or more random access memories (RAM) including static random access memory (SRAM) and/or dynamic random access memories (DRAM) for instance synchronous DRAM (SDRAM). The various controllers **214**, companion circuits **216**, volatile memories **218** and/or nonvolatile memories **220** may be communicatively coupled via one or more buses (only one shown) **222**, for instance instructions buses, data buses, address buses, power buses, etc.

The controllers **214** and/or companion circuitry **216** may execute instructions stored in or by the non-volatile memories **218** and/or volatile memories **220**. The controllers **214** and/or companion circuitry **216** may employ data, values, or other information stored in or by the volatile memories **220** and/or nonvolatile memories **218**.

In an embodiment of the present system and method, the control subsystem **212** may incorporate audio filtering circuitry or implement audio filtering by way of a general purpose processor which processes suitable instructions stored in non-volatile memory **218** or volatile memory **220**. Audio filtering may, for example, implement signal processing or data comparisons as described further herein to distinguish acceptable user speech from background user

speech. Audio filtering may rely upon a comparison of frames of speech provided from microphone **202**, via codec **208** and buffer **210**, with previously-established speech samples stored in nonvolatile memory **218** or volatile memory **220**.

In an alternative embodiment of the present system and method, some or all audio filtering, speech-processing, and speech-comparisons may be instead be accomplished via circuitry on the speech recognition device **106** (FIG. 1), **300** (FIG. 3). In an alternative embodiment, some or all audio filtering may be distributed between hardware and/or software of the headset **104**, **200**, and hardware and/or software of the speech recognition device **106**, **300**.

As described further herein below, in an embodiment of the present system and method, the sound signal from the microphone **202** is passed to the processor-based speech recognition device **106** (FIG. 1), **300** (FIG. 3) for speech recognition processing. The process of discriminating between user speech and background speech is then performed by the speech recognition device **106**, **300**.

The headset **200** optionally includes one or more radios **224** (only one shown) and associated antennas **226** (only one shown) operable to wirelessly communicatively couple the headset **200** to the processor-based speech recognition device **106** and/or backend computer system **110**. The radio **224** and antenna **226** may take a variety of forms, for example a wireless transmitter, wireless receiver, or wireless transceiver. In an embodiment where the headset **104**, **200** and SRD **106**, **300** are connected by a wired connection, radio **224** may not be required, or may be required only to communicate with the backend computer system **110**.

The radio **224** and antenna **226** may, for instance, be a radio suitable for short range communications, for example compatible or compliant with the Bluetooth protocol, which allows bi-directional communications (e.g., transmit, receive). Alternatively, the radio **224** and antenna **226** may take other forms, such as those compliant with one or more variants of the IEEE 802.11 protocols (e.g., 802.11n protocol, 802.11ac protocol). The radio **224** and antenna **226** may, for example, take the form of an RF communications card, received via a connector, for instance a PCMCIA slot, to couple the RF communications card to the controller **214**. RF communications cards are commercially available from a large number of vendors. The range of the radio **224** and antenna **226** should be sufficient to ensure wireless communications in the expected work environment, for instance wireless communications with a processor-based speech recognition device **106** worn by a same user as wears the headset **200**.

In an alternative embodiment, some or all of the electronic circuitry described above as being part of the headset **104**, **200** may instead be placed on the SRD **106**, **300**. The circuitry of the SRD **106**, **300** is discussed further immediately below.

Processor-Based Speech Recognition Device

FIG. 3 is a system diagram of an exemplary processor-based speech recognition device **300**, according to one embodiment of the present system and method. The processor-based speech recognition device **300** may be similar to or even identical to the processor-based speech recognition device **106** of FIG. 1.

The processor-based speech recognition device **300** may include one or more controllers, for example a microprocessor **302** and DSP **304**. While illustrated as a microprocessor **302** and a DSP **304**, the controller(s) may take a variety of forms, for instance one or more microcontrollers, ASICs, PGAs, GRUs, and/or PLCs.

The processor-based speech recognition device **300** may include one or more non-transitory computer- or processor-readable storage media such as non-volatile memory **306** and volatile memory **308**. Non-volatile memory **306** may take a variety of forms, for example one or more read-only memories (ROMs), one or more writeable memories, for instance EEPROM and/or one or more FLASH memories. The volatile memory **308** may take a variety of forms, for example one or more random access memories (RAM) including static and/or dynamic random access memories. The various controllers **302**, **304** and memories **306**, **308** may be communicatively coupled via one or more buses (only one shown) **310**, for instance instructions buses, data buses, address buses, power buses, etc.

The controllers **302**, **304** may execute instructions stored in or by the memories **306**, **308**. The controllers **302**, **304** may employ data, values, or other information stored in or by the memories **306**, **308**. The memories **306**, **308** may for example store instructions which implements the methods described further below herein to distinguish user speech from background speech, as in exemplary methods **400** and **600** (see FIGS. 4 and 6). The controllers **302**, **304**, when implementing these instructions, thereby enable the speech recognition device **300**, **106** to distinguish user speech from background speech.

The processor-based speech recognition device **300** optionally includes one or more radios **312** and associated antennas **314** (only one shown) operable to wirelessly communicatively couple the processor-based speech recognition device **300**, **106** to the headset **200**, **104**. Such radio **312** and antenna **314** may be particularly suited to relatively short-range communications (e.g., 1 meter, 3 meters, 10 meters). The radio **312** and antenna **314** may take a variety of forms, for example a wireless transmitter, wireless receiver, or wireless transceiver. The radio **312** and antenna **314** may, for instance, be a radio suitable for short range communications, for example compatible or compliant with the Bluetooth protocol. The range of the radio **312** and antenna **314** should be sufficient to ensure wireless communications in the expected work environment, for instance wireless communications with a processor-based headset **104**, **200**.

The processor-based speech recognition device **300** optionally includes one or more radios **316** and associated antennas **318** (only one shown) operable to wirelessly communicatively couple the processor-based speech recognition device **300**, **106** to the backend computer system/server **110** (FIG. 1), for example via one or more antennas **112** (FIG. 1) of a wireless network or communications system. The radio **316** and antenna **318** may take a variety of forms, for example a wireless transmitter, wireless receiver, or wireless transceiver. The radio **316** and antenna **318** may, for instance, be a radio suitable for relatively longer range communications (e.g., greater than 10 meters), for example compatible or compliant with one or more variants of the IEEE 802.11 protocols (e.g., 802.11n protocol, 802.11ac protocol) or WiFi protocol. In many applications, the range of the radio **316** and antenna **318** should be sufficient to ensure wireless communications in the expected work environment, for instance wireless communications with one or more antennas **112** (FIG. 1) positioned throughout the work environment, but this is not necessary.

General Speech Analysis Considerations

Note that the terms frames and fragments are used interchangeably throughout this specification to indicate information associated with a segment of audio. Also note that frames or fragments for the purposes of classification into user speech and background speech do not necessarily need

to correlate one to one to frames or fragments generated for purposes of feature generation for other aspects of speech recognition, e.g., speech detection, training, decoding, or general background noise removal. They may have many different parameters, such as using different frame rates, amounts of overlap, number of samples, etc.

A speech recognition system attempts to map spoken human speech to known language vocabulary. To do so, a voice system will, among other operational elements, typically compare (i) received real-time speech against (ii) a stored audio template, also referred to as an audio characterization model ACM, of previously captured/analyzed voice samples. Such an audio template is derived from a collection of voice training samples and other training samples referred to, for the present system and method, as the training corpus TC.

In general, speech recognition may involves several general stages. Presented here is an exemplary general process for real-time speech interpretation.

(1) Conversion of received sound to digital signal—Audio waves emanating from a human speaker, as well as nearby sounds from other sources, are converted to an analog electrical signal. This may be done for example by a microphone **120**, **202** in a headset **104**, **200**. The analog electrical signal is then digitalized, i.e., converted to binary 1's and 0's. This may be accomplished for example by the CODEC **208** of the headset **104**, **200**, or by the processor **302** of the speech recognition device **106**, **300**.

(2) Division of digitized sound into frames—The digitized sound is divided into frames, that is, segments of suitable length for analysis to identify speech. The length of segments may be geared to identify specific phonemes (sound units, such as a vowel sound or a consonant sound), or words or phrases.

NOTE: Further processing stages identified immediately below may be performed, for example, by the microprocessor **302** or digital signal processor **304** of the speech recognition device **106**, **300**, possibly based on instructions stored in non-volatile memory **306** or volatile memory **308**. In an alternative embodiment, these tasks may be performed in whole or part by elements of headset **104**, **200**, or server **110**.

(3) Conversion to frequency domain—The frames of the received, digitized audio signal are typically converted from the time domain to the frequency domain. This is accomplished for example via a Fourier transform or Fast Fourier transform, or similar processing.

(4) Conversion to secondary representation (state vectors)—In an embodiment, a frequency domain representation may be converted to other mathematical representations better suited for further processing. For example, while the frequency domain representation may be substantially continuous, various forms of concise representations may encapsulate the essential or key elements of the frequency domain representation. For example, amplitudes at various specific frequencies may be captured, or amplitudes of only the peak frequencies may be captured. Various other mathematical encapsulations are possible as well. The resulting mathematical characterization of the audio frames is sometimes referred to as “state vectors”.

(5) Normalizations and other supplemental signal processing—One of the challenges inherent in voice recognition is that human voices differ in their harmonics and speech patterns; for example, the same exact word spoken by two different persons may sound dramatically different in a variety of respects, such as pitch, loudness, and duration, as well as variations due to age, accents, etc. To help compensate for this, voice systems typically attempt to

normalize diverse samples of the same speech to similar mathematical representations. Thus, normalizations attempt to ensure that, for example, human vowel sounds (such as “ah”, “eh”, or “oh”) coming from different speakers will all have a substantially similar mathematical representation, common to all speakers, during processing. The process of converting digitized speech samples from different speakers to a partially or substantially similar form is referred to as “normalization.” A variety of established methods for this are known in the art.

In embodiments of the present system and method, one exemplary method of normalization is Vocal Length Tract Normalization (VTLN), which applies compensations for the varied pitches of the human voice (including, but not limited to, the typical differences between male and female voices). In alternative embodiments of the present system and method, another system of normalization which may be employed is Maximum Likelihood Linear Regression (MLLR), which adapts parameters within the stored template data to be a closer to match to a currently received sound signal.

Other signal conversions may be employed as well at various stages. For example, various frequency domains may be either boosted or suppressed.

(6) Comparison of received voice signal against the template—The processed, received voice signal is compared against a template of pre-processed, stored voice signals also referred to as an audio characterization model ACM. A favorable comparison is indicative of a user voice, which is accepted by the speech driven system **102**; an unfavorable comparison is indicative of a background voice (or possibly a user voice which is corrupted by extraneous background sounds), and which is thereby rejected by the voice driven system **102**.

Audio Characterization Model and Training Process

The audio characterization model ACM typically includes stored mathematical representations of human voices expressing certain words, for example storing the state vectors described above. The audio characterization model ACM also contains data which matches the stored audio signals to specific textual representations, i.e., textual transcriptions of the spoken words. The audio signal representations (state vectors) and textual transcriptions may be vowel or consonant sounds, whole words, phrases or sentence fragments, or even whole sentences. The comparison discussed above determines if the received voice signal is a match for a voice signal in the audio characterization model ACM (the stored audio template).

In an embodiment of the present system and method, a training corpus is prepared during a training phase which occurs in time prior to the release/use of the speech driven system **102** for field-use in factories, warehouses, or other industrial environments. The ACM (that is, the template) is prepared from the training corpus. Thus the audio characterization model ACM may be understood in part as a preset or pre-determined representation of correlations between audio signal representations (state vectors, etc.) and associated text, such as syllables, words, and/or phrases.

Training environment: In an embodiment of the present system and method, the audio vocabulary of the training corpus is initially recorded in a training environment which is the same as, or which mimics, an industrial environment in which the speech recognition device **106, 300** may be used. In this way, the audio samples in the training corpus are likely to be representative of audio samples which will be obtained from actual device users **100** during field operations. For example, if the speech recognition device

106, 300 is to be used in factory and warehouse settings, then the audio samples collected for training purposes may be collected in an actual factory or warehouse setting, or an environment designed to mimic such settings.

In one embodiment, the use of a field-realistic setting to record training sounds includes an environment which may have one or more of the following audio or acoustic aspects:

Background sounds from Public Address (PA) system voices. A factory, warehouse, or other industrial environment may include a public address system which delivers widely audible background speech at unpredictable times and/or of unpredictable content (for example, a generalized and unpredictable vocabulary).

Background voices from roving persons in the environment. In a factory, warehouse, or other industrial setting, the person using the speech recognition device **106, 300** will typically wear a headset **104** which is configured with a microphone **120** in immediate proximity to the user’s mouth (see FIG. 1 above). However, other persons working in the industrial setting may also be nearby, and the microphone **120** will pick up their speech as well.

Background sounds from operating equipment and machinery in the environment. For example, a factory environment may have sounds which include the operations of manufacturing machines and conveyor belts, while a warehouse environment may have sounds which include forklifts and other devices for movement of heavy objects.

It will be understood by persons skilled in the art that, as detected by the microphone **120**, background voices for example those from a PA system or from roving persons—will have audio qualities which are distinctive from the audio qualities of a user **100** whose mouth is in immediate proximity to the microphone **120**. Physiologically-based differences in the voices (between the user **100** and roving persons) also result in audio quality differences. It is a feature of the present system and method to distinguish a user voice from a background voice emitted from a PA system or from a roving person.

In an alternative embodiment, a training corpus may be obtained in an audio environment which is not the same as the field environment, for example, in a sound studio.

Training vocabulary: In an embodiment of the present system and method, the speech recognition device **106, 300** may be expected to be used principally in conjunction with a specific or well-defined vocabulary of terms. For example, it may be anticipated that users of the speech recognition device will principally speak terms associated with certain manufacturing processes or with certain warehouse procedures.

For example, the vocabulary may entail the use of digits or numbers, the names of certain specific procedures, and/or certain specific signal words or confirmation words for known tasks. In an embodiment, and in such cases, the vocabulary for the training corpus (and so, ultimately, for the audio characterization model ACM) may be principally confined to words, terms, or phrases which are expected/anticipated to be used by the users **100** of the speech recognition device **106, 300**.

In an alternative embodiment, the vocabulary for the training corpus may be a substantially more extended vocabulary, including terms and phrases of broader general usage apart from the most commonly expected terms for the particular field environment.

“Training users” and generalized user audio training: In an embodiment of the present system and method, the

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training corpus is representative of selected word sounds or word phrases, as they may be potentially by spoken by many different individuals. This may include individuals of different genders, different ages, different ethnic groups, persons with varying accents, and in general people whose widely varying physiologies may result in a broad array of distinctive vocal qualities, even when voicing the same word or phrase.

In an embodiment, this may entail that during the training phase, multiple different persons, referred to here as training users, are employed to help create the training corpus. In an embodiment, each such person (that is, each training person) is present in the training environment (not necessary at the time same time). It is noted that, while the training users could be the same as some people who will use the speech driven system 102 in the field, more typically the training users are not actual users 100. In an embodiment of the present system and method, the training users may be selected to represent various typical users or generic users 100. In an alternative embodiment, the training users may be selected to be representative of certain expected sub-populations of typical users 100, for example male users or female users.

During training, each training user dons a headset 104 with microphone 120 (see FIG. 1). With suitable prompts, each training user speaks a designated training vocabulary. Training users may be directed to recite the vocabulary, or parts of the vocabulary, multiple times and/or at multiple places within the training environment.

In this way, and for a single word or phoneme (e.g., “one”, “two”, “confirmed”, “stored”, etc.) multiple redundant samples may be gathered from each training user. Each such audio sample may sometimes be collected with just the word and no background sounds, and at other times with varying elements of background sounds (PA sounds, other roving speakers, machine sounds) from the training environment. In addition, the same training samples, provided by the multiple different training speakers, results in redundancy in terms of having multiple voice samples of the same text.

Combined user voices and background sounds: As will be apparent from the above description, the collected voice samples from training users may include background sounds. In an embodiment of the present system and method, training users may be deliberately directed to utter some voice samples when little or no background sounds are present; and to speak other voice samples (including possibly redundant trainings vocabulary) when background sounds are present. As a result, the training corpus may include the same training vocabulary both with and without background sounds.

Digitization and normalization of the training voice samples: After collection of the voice samples in the training environment, the voice samples are digitized and combined into the training corpus, for example in the form of their raw audio spectrum. The audio characterization model ACM may contain the same audio samples in compressed forms, for example in the form of state vectors, or in other signal representations.

The process of integrating the audio samples into the audio characterization model ACM may include various forms of mathematical processing of the samples such as vocal length tract normalization (VLTN) and/or maximum likelihood linear regression (MLLR). In an embodiment of the present system and method, the result is that within the audio characterization model ACM, a digitized samples of a single training word (e.g., “one,” “two,” “three,” etc.) may be normalized to represent a single, standardized user voice.

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In an alternative embodiment, within the training corpus a digitized sample of a single training word may be given multiple digital representations for multiple types of voices (for example, one representation for a generic female voice and one representation for a generic male voice).

In an embodiment of the present system and method, the audio characterization model ACM may include both one or more discrete samples of a given training word without background sounds; and one or more samples of the given training word with background sounds. In an alternative embodiment, the training corpus combines all instances of a training word into a single sample.

Transcribing and Marking the Training Corpus: In an embodiment of the present system and method, the digitized training corpus must be transcribed and tagged. This process entails have a training listener (or multiple training listeners) listen to the corpus. Each training listener is assigned to transcribe (via a keyboard action or mouse interaction with a computer, or similar) recognizable words or phrases, such as by typing the text of the recognizable words or phrases.

Further pre-field-use processing then includes combining the transcribed text digitally into the audio characterization model ACM. In this way, the finalized audio characterization model ACM includes both digital representations of recognizable audio; and, along with the digital representations, text of the associated word(s).

Acceptable and Unacceptable Articulations of Words: In addition, the training listener may be tasked to provide additional flags for sounds within the training corpus. This is also referred to as tagging. The present system and method pertains to distinguishing speech of a user of a voice recognition system from other speech which is background speech. In an embodiment of the present system and method, and during the training process, the training listener may flag (tag):

- (i) Words or phrases in the training corpus which are deemed to be of adequate audio clarity—that is, the words or phrases can be readily understood by the training listener with little or no uncertainty—which are flagged by the training listeners as acceptable. Typically, this reflects that the words were spoken by a training user, and that upon hearing, the articulation of the words is not obscured by background voices or sounds.
- (ii) Words or phrases in the training corpus which are deemed to be of poor audio clarity—that is, the words or phrases cannot be readily understood by the training listener, or can be only tentatively identified—and which are flagged by the training listeners as unacceptable. This typically reflects either (i) user speech which is obscured by excess background voices or other background sounds, or (ii) voices which are in fact background voices and not user speech.
- (iii) Background sounds associated with the industrial environment, such as machine noises or sounds from transport vehicles, which may also be flagged as unacceptable.

Finalized Audio Characterization Model ACM: The finalized audio characterization model ACM includes multiple elements, which may include for example and without limitation:

- (a) digital representations (e.g. state vectors) of audio based on user vocalizations for a selected vocabulary of words;
- (b) text of the words associated with the vocalizations;
- (c) digital representations of background sounds; and

(d) flags to distinguish acceptable audio samples from unacceptable audio samples.

The digital representations of training audio may include representations of user speech samples which were detected along with simultaneous background sounds, such as PA system voices, roving person voices, and other background sounds. These digital representations may therefore be indicative of user speech combined with expected background sounds for the industrial environment.

Audio Feature Extraction and Quantization: Sound Characterizations, Audio Characterization Models and Rejection Threshold: In an embodiment of the present system and method, a further stage of the pre-field-use process includes establishing a sound characterization for each speech audio sample in the audio characterization model. The sound characterization is indicative of a standardized sound quality of each audio sample, and in an embodiment may be derived via one or more mathematical algorithms from the spectrum of the audio sample. For example, the sound characterization may be based on a VTLN of each speech sample. In an alternative embodiment, the sound characterization may be based on an MLLR of each speech sample. The collection of sound characterizations and related threshold data (discussed below) constitute the audio characterization model ACM for the audio environment.

In an alternative embodiment, the sound characterization may be based on one or more formants, such as the lower order (1st, 2nd and/or 3rd) formants, of each speech sample; it may be based on raw values of the formants, normalized values, spacing between formants, or other related calculations. (Speech formants are either the spectral peaks of a sound and/or the resonances associated with the spectral peaks.)

In an alternative embodiment, the sound characterization for each speech audio sample is not determined during pre-field-use processing; rather, spectral data is stored directly in the audio characterization model, and the sound characterizations in the model are calculated by the speech recognition device 106 during run-time, that is, during field use.

In an embodiment of the present system and method, a final stage of the pre-field-use process may include establishing a rejection threshold. In an embodiment, the rejection threshold may be a specific mathematical value which distinguishes acceptable user speech from unacceptable user speech.

In an embodiment of the present system and method employment a neural network or other trained learning system, a final stage of the pre-field-use process may entail classifying the audio as one of "User Speech", "Background Speech", "PA Speech", with the possibility of also including a "Environment Noise" classification.

In embodiment of the present system and method, in field use, received vocalizations in the field (for example, in a factory or warehouse) may be processed by either headset 104, 200 or by speech recognition device 106, 300 to obtain a real-time sound characterization of the received vocalization. The sound characterization for the received speech may be compared against a sound characterization stored in the audio characterization model ACM. If the difference between the two values is less than the rejection threshold, the received vocalization is construed to be user speech, and is accepted. If the difference between the two values is greater than the rejection threshold, the received vocalization is construed to not be user speech and is rejected.

In an embodiment of the present system and method, for speech in the audio characterization model ACM, explicit

sound characterizations values and an explicit rejection threshold may be established during the pre-field-use processing.

Implicit Audio Characterizations Models, Implicit Rejection Threshold and Learning System Training: In an alternative embodiment of the present system and method, implicit values are used to characterize the training vocalizations, and for the rejection threshold.

In one exemplary embodiment, a machine learning system is trained to distinguish acceptable user speech from user speech that is not acceptable due to excessive background noise. The learning system may be hardware or software-based, and may include for example and without limitation: a neural network system, a support vector machine, or an inductive logic system. Other machine learning systems may be employed as well.

In one embodiment of the present system and method, machine learning such as neural network training may occur concurrently with transcription (as discussed above). In such an embodiment, the human trainer may be the same person or persons as the training listener. In an alternative embodiment, neural network training or other learning system training may be separate, pre-field-use process.

In general, machine learning entails presenting the learning system with data samples, and conclusions based on the data samples. The learning system then defines rules or other data structures which can substantially reproduce the same conclusions based on the same data samples.

For example, a neural network system may be presented with the training corpus, and asked to classify a newly presented voice sample against the training corpus as being either acceptable or unacceptable. The training process relies on a human trainer to define, in fact, the correct assessment (for example, the speech as being acceptable or unacceptable).

The machine learning system generates a provisional hypothesis, either that the newly presented voice sample is acceptable or unacceptable. The machine learning system presents the hypothesis (acceptable or not). The human trainer provides feedback to the learning system, either confirming the hypothesis (for example, that a voice sample which was predicted as being acceptable is in fact acceptable), or rejecting the hypothesis (for example, indicating that a voice sample which was predicted as being acceptable was, in fact, not acceptable).

Responsive to the feedback from the human trainer, the neural system modified internal data structures responsive to the feedback, and according to a suitable training/learning algorithm. For example, in the case of a neural network, the learning system modifies adaptive weights of neural links. Over time, with enough training, the result is a network which can significantly reproduce the desired outcomes as defined by the human trainer. For example, the learning system learns to distinguish acceptable user speech from unacceptable user speech, as it would be determined by a user.

As described above, a learning system therefore is trained to distinguish a level of difference between (i) a newly presented user voice sample and (ii) a training voice sample stored within the audio characterization model ACM. Beyond a certain level of difference, the user voice sample is deemed unacceptable. Therefore, in the finalized model of the neural network, there is at least an implicit rejection threshold beyond which received audio vocalizations are not acceptable. Similarly, the learning system establishes an implicit sound characterization for sound samples stored in

the audio characterization model ACM. In field use, at least an implicit sound characterization is formed for received audio as well.

Further below in this document, reference will generally be made to sound characterizations, rejection thresholds, and more generally to the audio characterization model ACM which characterizes acceptable speech for the audio environment. It will be understood that the sound characterizations and rejection thresholds may be either explicit values, or may be implicitly stored as distributed parameters or data structures in a learning system such as a suitably trained neural network.

Exemplary Pre-Field-Use Creation of Learning Corpus, Audio Characterization Model, and Rejection Threshold Via Learning Algorithm

FIG. 4 presents a flow chart of an exemplary method 400 for pre-field-use creation of a training corpus TC, audio characterization module ACM, and establishment of a rejection threshold RT. The method is operable on one or more computer systems, for example one which may be similar to server 110 of FIG. 1, though the actual computer(s) used for training is (are) likely to be other than the server 110 of FIG. 1. Multiple computers may be employed, for example at different stages of the learning/training process. (For example, one or more computers may be employed during the audio collection phases 405, 407 described below, while one or more other computers may be employed during the transcription phases 410, mixing phases 415, and other phases or procedures described below.)

Some of the steps shown in method 400 may be performed in different orders, or in parallel. The order of presentation below is for convenience only, and should not be construed as limiting. The method may be performed in part or in whole in a training environment, as described above.

The method may begin at step 405. In step 405, training users are prompted to voice speech samples, for example, to recite samples of a limited vocabulary which is expected to be employed in field-use of the speech driven system 102. The training users are typically wearing headsets 104, 200, with microphones suitably placed near their mouths, as described above. The speech samples are collected from the microphones 120, 202 of the headsets, and may be stored on a computer system as audio files (or in one integrated audio file) for further processing.

In an embodiment of the present system and method, user speech samples may be collected with no background sounds present. In an alternative embodiment, user speech samples may be collected when background sounds are always present (that is, audible and concurrent in time) as well. In an alternative embodiment, user speech sounds may be collected both with and without background sounds being present.

In step 407, audio samples are collected, via the user headsets 104, 200 as worn by the training users, of background sounds in the training environment. The background sounds emanate from other persons in the training environment, that is, persons other than the training user wearing the headset. The background voices may be recorded from persons at varying distances and in varying positions in relation to the user and the headset 104, 200.

In steps 410a and 410b (collectively, 410), the recordings of the audio samples are transcribed by the training listeners, including both transcriptions of user voice samples and transcriptions of background voice samples.

In step 415, some or all of the audio user speech samples may be mixed with audio of background speech samples. In

this way, audio representations may be created which were not actually heard in the training environment. For example, a single training word spoken by users (for example, a word such as “three” or “selected”) can be mixed with multiple different background sounds.

In an embodiment, multiple such samples may be created with, for example and without limitation: a single user word mixed with single different background sounds; a single user word mixed with multiple concurrent background sounds; and a single user word mixed with background sounds at different relative sound levels between the user sound and the background sound. In this way, many multiple realistic samples of user speech with background speech can be created from a limited set of samples initial.

In step 420, the present system and method calculates sound characterizations for the various combined user-voice/background-speech audio samples. As per the discussion above, the sound characterizations may be VTLNs of each sample, an MLLR of each sample, or other mathematical characterizations. In an embodiment of the present system and method, the VTLN values for users can be refined over time; as additional user voice samples are collected during field use (see FIGS. 5 and 6 below), the training corpus can be revised and redistributed to the various speech recognition devices 106 in the field.

In step 425, the present system and method collects the sound characterizations for the different phonemes or words, and tags them based on the transcriptions, thereby creating part of the training corpus TC. The tags may include the text representation associated with the audio or other text indicator (that is, an indicator of meaning) associated with the audio; and the tag may also include the determined quality indicator of “acceptable” or “unacceptable”. For some audio samples, the training listener may be unable to determine the spoken words, in which case only a quality tag or quality indicator may be provided.

Returning to step 405, where the speech samples were collected from training users, in step 430, the present system and method calculates sound characterizations for the various user voice samples. As per the discussion above, the sound characterizations may be VTLNs of each sample, an MLLR of each sample, or other characterizations.

In an embodiment of the present system and method, in step 430 it is possible to use VTLN factors calculated given various examples for a given user in place of the current VTLN factor for each example. This increases the number of examples by the number of pre-calculated VTLN factors used.

In step 435, the present system and method collects the sound characterizations for the different user phonemes or words, and tags them based on the transcriptions, thereby creating part of the training corpus TC. The tags may include the text representation associated with the audio or other text indicator (that is, an indicator of meaning) associated with the audio; and the tag may also include the determined quality indicator of “acceptable” or “unacceptable”.

In an embodiment where the user voice samples are collected without background sounds, it is expected that most or all of the user voice samples will be of sufficient clarity to be acceptable and to be transcribed. However, any such user voice samples which are not sufficiently clear may be tagged as unacceptable. In an embodiment where some or all user voice samples are collected with background sounds present, it is expected that at least some audio samples will not be fully intelligible, in which case only a quality tag or quality indicator may be provided.

The training corpus TC consists of all the tagged, transcribed audio samples, possibly with suitable condensation (for example, multiple samples of the same word or the same phoneme may be condensed to one representation).

In step 440, the present system and method determines a suitable audio characterization model ACM and Rejection Threshold RT. The audio characterization model ACM includes sound characterizations for multiple words, phrases, and/or phonemes. In an embodiment, and as described above, this may entail training a learning system to distinguish acceptable voice samples from unacceptable voice samples, and thereby having the learning system establish a suitable rejection threshold (either explicit or implicit).

Audio Environment in Field-Use

FIG. 5 illustrates an exemplary audio environment 500 for field use of a speech-driven system 102 including a headset 104, 200 and speech recognition device 106, 300. Such an environment may be present for example in a factory, a warehouse, or other industrial environment where the speech-driven system 102 may be employed.

The audio environment may include activity prompts 507 or other prompts which are provided to the user 100 from the SRD 106, 300 (sometimes via an application running on the SRD, and other times originating from server 110), and which the user 100 hears via headphones 118 of headset 104. These prompts may include instructions for field activity, such as selecting certain items from specified locations or bins in a warehouse, or may include prompts for certain manufacturing activities such as stages in assembly of some hardware. The prompts may include prompts to the user to speak certain words specifically for audio-training purposes (see for example step 602 of method 600, FIG. 6 below).

In response to prompts, the user 100 may either speak certain expected, prompted responses 505a, or perform certain activities, or both. For example, a prompt 507 may tell a user to pick an item from a bin numbered "A991". In response, the user may be expected to recite back the words "'A' nine nine one", then actually pick the item from a bin numbered "A991", and then recite some confirmation phrase such as "picked" or "selected."

Hinted user speech and non-hinted user speech—In general, the user's speech 505 comprises both: (i) some speech in response to prompts, where a specific response or specific choices of response to the prompt is/are expected from the user, and which is referred to as hinted speech 505a; and (ii) all other user speech 505b, which is non-hinted user speech. Some prompted responses are hinted, which occurs in specific parts of the user's workflow (for example, when the user is prompted to select a particular part from storage). Hinted speech will have an expected value for a reply, which is typically a dynamic value that is associated with the task at hand (for example, a part number).

Non-hinted speech 505b may include general conversation which the user engages in with other persons, as well as some requests or other data provided to the server 110 by the user 100. All user speech 505 is detected by microphone 120 of headset 104.

Speech detected by microphone 120, 105 in the field will also typically include background speech 510 from other persons in the area, and PA system speech 515.

Stored data on speech recognition device: In embodiment of the present system and method, all collected sounds—user speech 505, background speech 510, and PA system speech 515, as well as other background sounds (not illus-

trated) are transmitted or passed from headset 104 as audio samples 520 to the speech recognition device (SRD) 106, 300.

In an embodiment of the present system and method, the SRD 106, 300 is pre-programmed with, pre-configured with, and/or stores both a suitable audio characterization model ACM and/or training corpus TC for the current industrial environment, and a suitable rejection threshold TC.

In an embodiment, SRD 106, 300 is also pre-programmed with, pre-configured with, and/or stores a vocabulary of hint text HT expected to be used in the field. In an alternative embodiment, some or all of the audio characterization model ACM, training corpus TC, rejection threshold RT, and/or Hint Text HT may be prepared or stored on server 110 (for example, at the factory or warehouse where the SRD is to be used).

Exemplary Method of Field-Use of the Speech Recognition Device

FIG. 6 illustrates an exemplary method 600 of field-use of a speech-driven system 102. In an embodiment, the method is operable in a warehouse, factory, or other industrial setting with an audio environment the same or similar to exemplary field-use audio environment 500 (see FIG. 5 above).

The method 600 begins with step 602, which entails training the speech recognition device (SRD) 106, 300 to be operable with a particular user 100, by recognizing the voice of the user 100. In an embodiment, step 602 is a field-use step, but is a one-time step and is performed preliminary to the main use of the SRD 106, 300 to support user activity in the field.

In an embodiment, the training of the speech recognition device may entail prompting the user to speak specific, expected words, typically a limited vocabulary of words which the user will employ in the course of work. These prompted words may include digits, numbers, letters of the alphabet, and certain key words which may be commonly used in a given setting, for example, "Ready", "Okay", "Check", "Found", "Identified", "Loaded", "Stored", "Completed", or other words which may indicate a status of a warehouse or factory activity. In an embodiment, the prompted words may include some or all of the words expected to be used as hint words 505a during field use (see FIG. 5 above).

The user is prompted to speak these words (typically one word at a time), and the user then speaks the prompted words in reply. The SRD 106, 300 records the user replies and digitizes them. In an embodiment, the SRD 106, 300 may calculate for each word a set of state vectors and/or sound characterizations, employing calculations in a manner similar to that of pre-field use processing (see FIG. 4 above).

In an embodiment of the present system and method, state vectors and/or sound characterizations obtained during training step 602 may be used to modify the current user characterization and/or the rejection threshold RT which is stored on the SRD 106, 300.

Routine or on-going field-use of the SRD may commence with step 605. In step 605, the headset 104 functions interactively with the user 100 and the larger audio environment 500. Prompts 507 for user activity may be provided via headphones 104. The prompts 507 may originate, for example, on server 110. Responsive to the prompts 507, the user 100 may engage in various appropriate activities, such as for example picking stock items from prompted locations, moving stock items to prompted locations, or other activities. Responsive to the prompts and to their own activity, the user may also speak various words 505. These user-spoken words 505 or phrases may for example confirm recognition

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of a prompt, or may confirm completion of a task, or may confirm identification of a location or object.

The user spoken words **505** are detected by microphone **120**, **200** of headset **104**. Microphone **120**, **200** may also detect background speech **510** from other persons present in the environment, as well as PA system speech **515**, and other background sounds.

In an embodiment, step **605** may entail creation of digitized audio samples, packets, or frames **520**, which may include user speech **505**, background speech **510**, PA system speech **515**, and other sounds, either concurrently or serially. In an embodiment, the digitized audio samples **520** are passed from headset **104** to SRD **106**.

In step **610** the SRD **106** calculates a suitable sound characterization **690** of the audio sample **520**. Suitable sound characterizations are those which are comparable to those stored in the training corpus TC. For example, suitable sound characterizations may include VTLNs of an audio sample **520**, or MLLRs of an audio sample **520**. Other sound characterizations, suitable to match those of training corpus TC, may be employed as well.

Comparison of Received Sound with Hint Text: In one embodiment of the present system and method, in step **615** the method compares the received audio sample **520**, and/or the sound characterization of the received audio sample **690**, against stored sound characterizations of the hint text HT.

In step **620**, a determination is made if the received audio sample **520**, **690** matches any of the words in the hint text HT. If the received audio sample **520**, **690** matches hint text HT, then it is presumed that the audio sample **520**, **690** comes from a valid user **100**, and further processing with respect to possible background speech may be skipped. In this event, the method proceeds to step **625** (along the path marked “Yes” in the figure), where the received audio sample **520**, **690** is accepted.

In an embodiment of the present system and method, the SRD **106**, **300** may also at some times be running in a field training state or a noise sampling state. In such a training or noise-sampling state, then at step **620** (and whether or not the received audio sample **520** matched the hint text HT) the method would automatically accept the user speech; the method would then automatically proceed to steps **625** and **630**, discussed immediately below.

In an embodiment of the present system and method, from step **625** the method may proceed to step **630**. In step **630**, the present system and method uses calculated sound characterization **690** to improve the stored user sound characterizations and/or the audio characterization model ACM. For example, the state vectors which characterize user speech in the training corpus TC may be refined based on actual speech from actual users in the field. In an embodiment, this refinement of the training corpus TC may occur in the field in substantially real-time, with the sound characterizations in the training corpus TC being updated in real-time.

In an embodiment of the present system and method, each time the user **100** powers up the SRD **106**, **300**, the system starts over in building the characterization for that specific user. In an alternative embodiment, the SRD **106**, **300** may persist the user characterization across power cycles (for example, storing the characterization in memory **306**), so it is not necessary to start over each time. This specific characterization would not be stored in the audio characterization model (ACM).

In an alternative embodiment, user sound characterizations **690** collected in the field may be stored on server **110** or another suitable storage medium; the collected user sound

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characterizations **690** may then be processed en masse, using methods the same or similar to those of method **400** (FIG. 4, above).

Returning now to step **620**, it may be the case that the received audio sample **520** and/or the sound characterization **690** of the received speech does not match any of the vocabulary in the stored list of hint text HT. In that event, the method continues with step **635** (along the path marked “No” in the figure).

In step **635**, the present system and method compares the sound characterization **690** of the received audio sample against sound characterizations in the audio characterization model ACM. The comparison searches for an acceptably close match, but also determines a quality level of the match.

In step **640**, a determination is made as to whether the match is of acceptable quality. If the match is of acceptable quality (the path marked “Yes”), then in step **645** the speech is accepted as user speech. If the match is not of acceptable quality (the path marked “No”), then in step **650** the speech is rejected as not being user speech. As described above, in an embodiment such a determination made be made by a suitably trained learning system such as a neural network system trained as described above in this document (see for example FIG. 4 and associated discussion).

Shown in the figure is a supplemental example SE which illustrates one possible particular embodiment of steps **635** through **650**. In step **635E** (corresponding to step **635**), a difference value DV is calculated as the absolute value of the difference between the VTLN of the received audio sample **690** and the VTLN factor of a suitable audio example in the training corpus TC.

In step **640E** (corresponding to step **640**), a determination is made as to whether the difference value DV is less than the rejection threshold RT. If the match is of acceptable quality (so that the difference value DV is less than the rejection threshold RT), then in step **645E** the speech is accepted as user speech. If the match is not of acceptable quality (so that the difference value DV is greater than the rejection threshold RT), then in step **650E** the speech is rejected as not being user speech.

As will be appreciated by persons skilled in the art, once an audio sample has been approved as acceptable speech, the meaning of the audio sample may be determined (based for example on transcription data in the training corpus TC). Based on the meaning of the received audio sample, suitable further actions may be taken by the speech driven system **102**.

In an alternative embodiment of the present system and method, steps **615** and **620** (pertaining to the hint text comparison) may be omitted, along with omission of steps **625** and **630**. In such an embodiment, control may pass directly from step **610** to step **635**, **635E**.

SUMMARY AND ALTERNATIVE EMBODIMENTS

In an embodiment of the present system and method, a comparison is made between a real-time speech sample and pre-established sound samples. The pre-established sound samples are indicative of acceptable user vocalizations, and also indicative of unacceptable vocalizations—that is, vocalizations which are due to background voices, PA systems, or due to user vocalizations but which may be unintelligible due to concurrent background sounds.

A suitable metric is defined to analytically or numerically characterize the sameness or difference of the real-time speech sample against the pre-established sound samples.

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If a sound property of a real-time speech sample is sufficiently close or similar to the sound properties of the pre-established samples which are indicative of user voices, the real time-speech sample is categorized as being acceptable user speech.

If the sound property of the real-time speech sample is not sufficiently close or not sufficiently similar to the sound properties of the pre-established samples which are indicative of user voices, the real time-speech sample is categorized as being unacceptable background speech.

The level of closeness or difference between a real-time sound sample and the stored sound samples is determined with relation to a suitable threshold value.

Audio Comparison Matrix: In an embodiment, to distinguish user speech from background speech, the present system and method may employ a stored, audio-derived data structure which incorporates sound data as a basis for comparisons. In one embodiment, the audio data structure may be a sound matrix, or an array of sound characterizations. Some cells in the audio matrix tend to characterize sounds which are valid user voice sounds, while other audio matrix cells tend to characterize sounds which are background voice sounds.

In real-time, newly recorded sounds are compared against cells in the audio matrix. Incoming vocalizations which compare favorably with valid user vocalizations in the sound matrix are considered to be acceptable user speech; incoming vocalizations which do not compare favorably with valid vocalizations in the matrix are rejected.

In some embodiments, the number of different available, stored voice characterizations may be too many to store in a matrix or array; or the voice characterizations may blend to a degree that does not lend towards distinguishing the characterizations as discrete, one-cell-per-one sound characterization storage. Instead, other comparison methods, based on mathematically continuous representations of voice characterizations, may be employed.

Thus, in various embodiments, signal matching and comparison methods may employ other data structures than a matrix of sound characterizations to make a comparison. A variety of signal processing techniques and artificial intelligence techniques, including neural networks and other learning system techniques, may be used to compare real-time field vocalizations against data stored in distributed or other forms in the learning system.

FURTHER EMBODIMENTS

In further embodiments, labeled A1 through A10, the present system and method may also be characterized as:

A1: A speech recognition device (SRD) configured for recognition of human speech, comprising:

a microphone for receiving audio input;

a memory; and

a hardware processor communicatively coupled with said microphone and said memory, wherein said hardware processor is configured to:

identify in said received audio input a vocalization of a human language; and

categorize said received vocalization of a human language as either one of:

a first vocalization originating from a user of the SRD; or

a second vocalization originating from a background speech in the vicinity of the SRD.

A2: The SRD of embodiment A1, wherein:

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said memory is configured to store an audio characterization

model comprising:

a plurality of user speech samples; and

a background speech sample; and

said hardware processor is configured to categorize said received vocalization based on a comparison of the received vocalization with the stored audio characterization model.

A3: The SRD of embodiment A2, wherein:

the stored audio characterization model further comprises an audio mix with concurrent sound of:

a user speech sample of the plurality of user speech samples; and

the background speech sample; and

said hardware processor is configured to categorize said received vocalization based on a comparison of said received vocalization with said audio characterization model.

A4: The SRD of embodiment A3, wherein the memory is configured to store a speech rejection threshold; and the hardware processor is configured to:

compute a speech difference between the received vocalization and the audio characterization model;

categorize the received vocalization as the first vocalization originating from the user of the SRD if an absolute value of the speech difference is less than the speech rejection threshold; and

categorize the received vocalization as the second vocalization originating from a background speech if the absolute value of the speech difference is greater than the speech rejection threshold.

A5: The SRD of embodiment A4, wherein said speech rejection threshold comprises a pre-determined threshold calculated based on at least one of:

training samples of user speech; and

training samples of background speech.

A6: The SRD of embodiment A5, wherein said speech rejection threshold is dynamically updated in field-use based upon the first vocalization originating from the user of the SRD.

A7: The SRD of embodiment A2, wherein:

the memory is configured to store as the audio characterization model a normalized user speech;

the hardware processor is configured to normalize the received vocalization; and

the comparison comprises a comparison of the normalized user speech and the normalized received vocalization.

A8: The SRD of embodiment A1, wherein said hardware processor is configured to:

determine if the received vocalization matches an expected verbalization from among a vocabulary of one or more expected verbalizations;

upon determining that the received vocalization matches the expected verbalization, categorize said received vocalization as the first vocalization originating from the user of the SRD; and

upon determining that the received vocalization does not match any of the one or more expected verbalizations, categorize said received vocalization based on a comparison of the received vocalization with a stored characterization of user speech for the SRD.

A9: The SRD of embodiment A8, wherein a vocabulary of one or more stored expected verbalizations comprises a voiced word sample collected from the SRD user in the field.

A10: The SRD of embodiment A1, further comprising a pre-trained learning system, wherein said pre-trained learning system comprises at least one of:

a set of rules determined in training to distinguish user speech from background speech, based on a data set of training user speech and training background speech; and

a set of weighted connections of a neural network system, said weighted connections determined in training to distinguish user speech from background speech based on the data set of training user speech and training background speech;

wherein said pre-trained learning system is configured to process said received audio input according to the at least one of the set of rules and the set of weighted connections, and

wherein said pre-trained learning system categorizes said received vocalization as either one of the first vocalization or the second vocalization.

Persons skilled in the relevant arts will recognize that various elements of embodiments A1 through A10 can be combined with each other, as well as combined with elements of other embodiments disclosed throughout this application, to create still further embodiments consistent with the present system and method.

To supplement the present disclosure, this application incorporates entirely by reference the following commonly assigned patents, patent application publications, and patent applications:

U.S. Pat. No. 6,832,725; U.S. Pat. No. 7,128,266;
U.S. Pat. No. 7,159,783; U.S. Pat. No. 7,413,127;
U.S. Pat. No. 7,726,575; U.S. Pat. No. 8,294,969;
U.S. Pat. No. 8,317,105; U.S. Pat. No. 8,322,622;
U.S. Pat. No. 8,366,005; U.S. Pat. No. 8,371,507;
U.S. Pat. No. 8,376,233; U.S. Pat. No. 8,381,979;
U.S. Pat. No. 8,390,909; U.S. Pat. No. 8,408,464;
U.S. Pat. No. 8,408,468; U.S. Pat. No. 8,408,469;
U.S. Pat. No. 8,424,768; U.S. Pat. No. 8,448,863;
U.S. Pat. No. 8,457,013; U.S. Pat. No. 8,459,557;
U.S. Pat. No. 8,469,272; U.S. Pat. No. 8,474,712;
U.S. Pat. No. 8,479,992; U.S. Pat. No. 8,490,877;
U.S. Pat. No. 8,517,271; U.S. Pat. No. 8,523,076;
U.S. Pat. No. 8,528,818; U.S. Pat. No. 8,544,737;
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U.S. Pat. No. 8,559,767; U.S. Pat. No. 8,599,957;
U.S. Pat. No. 8,561,895; U.S. Pat. No. 8,561,903;
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U.S. Pat. No. 8,616,454; U.S. Pat. No. 8,621,123;
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U.S. Pat. No. 8,807,431; U.S. Pat. No. 8,807,432;
U.S. Pat. No. 8,820,630; U.S. Pat. No. 8,822,848;
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U.S. Pat. No. 8,842,849; U.S. Pat. No. 8,844,822;
U.S. Pat. No. 8,844,823; U.S. Pat. No. 8,849,019;
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In the specification and/or figures, typical embodiments of the invention have been disclosed. The present invention is not limited to such exemplary embodiments. The use of the term "and/or" includes any and all combinations of one or more of the associated listed items. The figures are schematic representations and so are not necessarily drawn to scale. Unless otherwise noted, specific terms have been used in a generic and descriptive sense and not for purposes of limitation.

The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flow charts, schematics, exemplary data structures, and examples. Insofar as such block diagrams, flow charts, schematics, exemplary data structures, and examples contain one or more functions and/or operations, it will be understood by those skilled in the art that each function and/or operation within such block diagrams, flowcharts, schematics, exemplary data structures, or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof.

In one embodiment, the present subject matter may be implemented via Application Specific Integrated Circuits (ASICs). However, those skilled in the art will recognize that the embodiments disclosed herein, in whole or in part, can be equivalently implemented in standard integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more controllers (e.g., microcontrollers) as one or more programs running on one or more processors (e.g., microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of ordinary skill in the art in light of this disclosure.

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In addition, those skilled in the art will appreciate that the control mechanisms taught herein are capable of being distributed as a program product in a variety of tangible forms, and that an illustrative embodiment applies equally regardless of the particular type of tangible instruction bearing media used to actually carry out the distribution. Examples of tangible instruction bearing media include, but are not limited to, the following: recordable type media such as floppy disks, hard disk drives, CD ROMs, digital tape, flash drives, and computer memory.

The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the present systems and methods in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the invention to the specific embodiments disclosed in the specification and the claims, but should be construed to include all voice-recognition systems that read in accordance with the claims. Accordingly, the invention is not limited by the disclosure, but instead its scope is to be determined entirely by the following claims.

The invention claimed is:

1. A method of speech recognition for use in warehouse picking operations, the method comprising:

receiving, at a microphone of a speech cognition device (SRD), an audio input, wherein the audio input is related to a confirmation phrase uttered in response to an instruction to pick an item;

identifying, via a hardware processor of the SRD communicatively coupled with the microphone, a vocalization in the received audio input;

classifying, via the hardware processor, a portion of the identified vocalization based on a machine learning model trained to understand input natural language, wherein the portion of the identified vocalization is classified as at least one of a user speech or a background speech;

in an instance in which the portion of the identified vocalization is classified as the user speech;

processing, via the hardware processor the portion of the identified vocalization, and

generating, via the hardware processor, at least one of words or phrases related to the warehouse picking operations, and

in an instance in which the portion of the identified vocalization is classified as the background speech, filtering via the hardware processor the portion of the identified vocalization.

2. The method of claim 1, wherein the machine learning model is at least one of a neural network system, a support vector machine, or an inductive logic system.

3. The method of claim 1, wherein the confirmation phrase is at least one of a confirmation of a recognition of a prompt, a confirmation of a completion of a task, or a confirmation of an identification of at least of a location and an object.

4. The method of claim 1, wherein the microphone is mounted on a headset and is configured for use in a warehouse environment.

5. The method of claim 1, wherein the machine learning model is trained based on a training corpus comprising at least one word or phrase related to the warehouse picking operations.

6. The method of claim 5, wherein the machine learning model is trained based on the training corpus and the audio input received during the warehouse picking operations.

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7. A speech recognition device (SRD) for use in warehouse picking operations comprising:

a microphone;

at least one processor; and

at least one memory including computer program code, the at least one memory and the computer program code configured to, with the at least one processor, cause the SRD to at least:

receive, at the microphone of the SRD, an audio input, wherein the audio input is related to a confirmation phrase uttered in response to an instruction to pick an item;

identify a vocalization in the received audio input;

classify a portion of the identified vocalization based on a machine learning model trained to understand input natural language, wherein the portion of the identified vocalization is classified as at least one of a user speech or a background speech;

in an instance in which the portion of the identified vocalization is classified as the user speech;

process the portion of the identified vocalization, and generate at least one of words or phrases related to the warehouse picking operations; and

in an instance in which the portion of the identified vocalization is classified as the background speech, filter the portion of the identified vocalization.

8. The SRD of claim 7, wherein the machine learning model is at least one of a neural network system, a support vector machine, or an inductive logic system and wherein the machine learning model is trained to distinguish the user speech from the background speech.

9. The SRD of claim 7, wherein the confirmation phrase is at least one of a confirmation of a recognition of a prompt, a confirmation of a completion of a task, or a confirmation of an identification of at least of a location and an object.

10. The SRD of claim 7, further comprising a headset, wherein the microphone is mounted on the headset and is configured for use in a warehouse environment.

11. The SRD of claim 7, wherein the machine learning model is trained based on a training corpus comprising at least one word or phrase related to the warehouse picking operations.

12. The SRD of claim 11, wherein the machine learning model is trained based on the training corpus and the audio input received during the warehouse picking operations.

13. A method of speech recognition for use in warehouse picking operations, the method comprising:

receiving, at a microphone of a speech recognition device (SRD), an audio input, wherein at least a portion of the audio input is related to a confirmation phrase uttered in response to an instruction to pick an item;

identifying, via a hardware processor of the SRD communicatively coupled with the microphone, a vocalization in the received audio input;

classifying, via the hardware processor, the portion of the identified vocalization based on a machine learning model trained to understand input natural language, wherein the portion of the identified vocalization is classified as at least one of a user speech, an environmental noise, or a background speech;

in an instance in which the portion of the identified vocalization is classified as the user speech;

processing, via the hardware processor, the portion of the identified vocalization, and

generating, via the hardware processor, at least one of words or phrases related to the warehouse picking operations;

in an instance in which the portion of the identified vocalization is classified as the background speech, filtering, via the hardware processor, the portion of the identified vocalization; and

in an instance in which the portion of the identified 5 vocalization is classified as the environmental noise, filtering, via the hardware processor, the portion of the identified vocalization.

14. The method of claim **13**, wherein the machine learning model is at least one of a neural network system, a support 10 vector machine, or an inductive logic system.

15. The method of claim **13**, wherein the confirmation phrase is at least one of a confirmation of a recognition of a prompt, a confirmation of a completion of a task, or a confirmation of an identification of at least of a location and 15 an object.

16. The method of claim **13**, wherein the microphone is mounted on a headset and is configured for use in a warehouse environment.

17. The method of claim **13**, wherein the environmental 20 noise comprises at least one of an operation of a vehicle in a warehouse or a movement of pallets in the warehouse.

18. The method of claim **13**, wherein the machine learning model is trained based on a training corpus comprising at least one word or phrase related to the warehouse picking 25 operations.

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